

Drainage Plan Drainage Strategy

22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP SURFACE & FOUL WATER DRAINAGE DESIGN Client – Sonila Lamaj COPYRIGHT THIS DOCUMENT MUST NOT BE COPIED OR REPRODUCED WITHOUT THE PRIOR PERMISSION OF J ROBERTS DESIGN LIMITED <redacted_url> 4555 / <redacted_date> Project 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr, ZZ29 6EP Doncaster Job Ref. 4555 Section SW + FW Drainage Design Report Sheet no./rev. 1 Calc. by LM Date <redacted_date> Chk'd by CAH Date <redacted_date> App'd by Date Introduction The following report and calculations are in justification of the surface and foul water solutions for a dwellinghouse. In accordance with Building Regulations <redacted_date> Part H3 surface water from a development should discharge to one of the following in priority order: ■ Soakaway or Infiltration System ■ Watercourse ■ Public Sewer Percolation testing was carried out by G&M Consulting Ltd (Appendix B). Due to clayey ground and low infiltration rates, it is not possible to use soakaways. There are no watercourses to discharge into as the site is landlocked. It is therefore proposed to discharge the surface water sewer network at 2.0l/s, located within 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP. The drained roof area is 300m² + 10% urban creep = 330m² (0.033Ha). The drained area of permeable paving is 170m². Overall the total trained area is <redacted_url> design standard for the attenuation tank is to not flood in the 1 in 100yr + 40% climate change return period. It is proposed to install an attenuation tank measuring 6.0m x 12.0m x 0.4m dp (72m² on plan). Foul water will connect to the public sewer network via gravity. Refer to Appendix A for the Drainage Layout and Details. Refer to Appendix B for Microdrainage Calculations + SILTBLOK Testing Efficiencies. Refer to Appendix C for Yorkshire Water Sewer Maps. Project 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr, ZZ29 6EP Doncaster Job Ref. 4555 Section SW + FW Drainage Design Report Sheet no./rev. 2 Calc. by LM Date <redacted_date> Chk'd by CAH Date <redacted_date> App'd by Date Standards and References BRE Digest 365 Soakaway Design BS EN 12056-2:<redacted_date> Building Regulations Part H Software MicroDrainage Design information to be read in conjunction with J Roberts Design Ltd General Structural and Foundation notes, Construction (Design and Management) Regulations <redacted_date>, CDM <redacted_date> – Domestic Clients: Roles and responsibilities, The Buildings Regulations <redacted_date> - Part A Project 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr, ZZ29 6EP Doncaster Job Ref. 4555 Section SW + FW Drainage Design Report Sheet no./rev. 3 Calc. by LM Date <redacted_date> Chk'd by CAH Date <redacted_date> App'd by Date Construction, Design and Management (CDM) J Roberts Design Ltd have undertaken the role of Designer for this project. It is assumed that an Architect has been appointed to the role of Principal Designer and has advised the Client of their responsibilities under the act. If an Architect has not been appointed, the roles and responsibilities under the CDM regulations normally pass to the principal contractor. Under CDM <redacted_date> it is our duty as designers to provide adequate information about any significant risks associated with the design project. Our scope of work on this project is: ■ Surface Water Drainage ■ Foul Water Drainage The concept and form of construction of the proposed work is conventional. In our opinion, the design solution reduces foreseeable risk to an acceptable level and no significant risks remain from the design in the construction or use that could reasonably have been avoided. We believe that the level of residual risk is normal for this type of project and would be obvious to a competent contractor. Additional information regarding the responsibility of a client commissioning domestic building work under the CDM regulations <redacted_date> can be found on the HSE website: <redacted_url> Project 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr, ZZ29 6EP Doncaster Job Ref. 4555 Section SW + FW Drainage Design Report Sheet no./rev. 4 Calc. by LM Date <redacted_date> Chk'd by CAH Date <redacted_date> App'd by Date SURFACE WATER DRAINAGE DESIGN SuDS CONSIDERATION SuDS Hierarchy Surface water drainage should be managed in a way that replicates the natural drainage processes for the site as closely as possible. The proposals should follow the NPPF hierarchy and should be disposed of to a receptor in the order of preference described below: 1. Into the ground; 2. To a surface water body e.g. watercourse; 3. To a surface water, highway drain, or another drainage system; 4. To a combined sewer. SuDS to be Considered In considering the appropriate SuDs components for the site Table 7.1 of CIRIA C753 indicates the benefits the components provide. The SuDs suitable for this site have been highlighted. Component Type Description Collection Mechanism Design Criteria Water Quantity Peak Runoff RatesWater QualityAmenityBiodiversity Runoff Volumes SmallEvents(Intercept LargeEvents Rainwater Harvesting Systems that collect runoff from the roof of a building or other paved surface for use. P ■ ■ ■ ■ Green Roofs Planted soil layers on the roof of buildings that slow and store runoff. S ■ ■ ■ ■ ■ Infiltration Systems Systems that collect and store runoff, allowing it to infiltrate into the ground. P ■ ■ ■ ■ ■ Proprietary Treatment Systems Subsurface structures designed to provide treatment of runoff. P ■ Filter Strips Grass strips that promote sedimentation and filtration as runoff is conveyed over the surface. L ■ ■ ■ ■ ■ Filter Drains Shallow stone-filled trenches that provide attenuation, conveyance and treatment of runoff. L ■ ■ ■ ■ ■ Swales Vegetated channels (sometimes planted) used to convey and treat runoff. L ■ ■ ■ ■ ■ ■ ■ Bioretention Systems Shallow landscaped

depressions that allow runoff to pond temporarily on the surface, before filtering through P ■■■■■■ Trees
 Trees with soil-filled tree pits, tree planters or structural soils used to collect, store and treat runoff. P ■■■■■
 ■ Pervious Pavements Structural Paving through which runoff can soak and subsequently be stored in the
 sub-base beneath, S ■■■■■■ Project 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr,
 ZZ29 6EP Doncaster Job Ref. 4555 Section SW + FW Drainage Design Report Sheet no./rev. 5 Calc. by LM
 Date <redacted_date> Chk'd by CAH Date <redacted_date> App'd by Date Attenuation Storage Tanks Large,
 below-ground voided spaces used to temporarily store run-off before infiltration, controlled release P ■
 Detention Basins Vegetated depressions that store and treat runoff. P ■■■■■■ Ponds and Wetlands
 Permanent pools of water used to facilitate treatment or runoff – runoff can also be stored in an P ■■■■■
 Key P – Point, L-Lateral, S-Surface ■ Likely valuable contribution to delivery design criterion o Some Potential
 contribution to delivery of design criterion, if specifically included in the design Project 22 Market Lane,
 Bessacarr, Doncaster, ZZ29 6EP, Bessacarr, ZZ29 6EP Doncaster Job Ref. 4555 Section SW + FW Drainage
 Design Report Sheet no./rev. 6 Calc. by LM Date <redacted_date> Chk'd by CAH Date <redacted_date>
 App'd by Date Surface Water Drainage Design Percolation testing was carried out by G&M Consulting Ltd
 (Appendix B). Infiltration into the ground via soakaways is not possible given the clayey ground and low
 infiltration rate. There are no watercourses to discharge into as the site is landlocked. It is therefore proposed
 to discharge roof run-off to the surface water sewer network at 2.0l/s, located within 22 Market Lane,
 Bessacarr, Doncaster, ZZ29 6EP, pending a S106 agreement with Yorkshire Water. The drained roof area is
 300m² + 10% urban creep = 330m² = 0.033ha. The area of permeable paving draining to the attenuation tank
 is 170m². The total drained area is 500m² or 0.050ha. The design standard for the attenuation tank is to not
 flood in the 1 in 100yr + 40% climate change return period. It is proposed to install an attenuation tank
 measuring 6.0m x 12.0m x 0.4m dp. The permeable paving will serve its own catchment and will drain to the
 surface water drainage system via 100Ø perforated pipes. The permeable paving will intercept the first 5mm of
 rain and also offer bioremediation. The fact that it will store some water has not been factored into the sizing of
 the attenuation tank, hence the system is more conservative. The storage will be offline so that during normal
 rainfall conditions, normal flow does not enter the attenuation tank, and thus, siltation is minimised, reducing
 the maintenance required. Given the small catchment restricting the flow to 2.0l/s/ha would require a very
 small opening in the flow control which would be prone to blockages. Therefore, a minimum discharge rate of
 2.0l/s has been used. Rainwater harvesting is not considered within the following design as it may be removed
 or not maintained over the life of the development, removing any benefit. Rainwater harvesting may be
 something the individual property owners wish to install but is not included within this design. A Microdrainage
 model was built to represent the proposed system (See Appendix B) and with an attenuation tank measuring
 6.0m x 12.0m x 0.4m. There will be extra capacity in the pipework etc. No flooding will occur for the 1 in 100y
 +40%CC scenario. Refer to Appendix A for the private drainage layout and details. Project 22 Market Lane,
 Bessacarr, Doncaster, ZZ29 6EP, Bessacarr, ZZ29 6EP Doncaster Job Ref. 4555 Section SW + FW Drainage
 Design Report Sheet no./rev. 7 Calc. by LM Date <redacted_date> Chk'd by CAH Date <redacted_date>
 App'd by Date Attenuation Tank Roof Area = 300m² + 10% urban creep = 330m² = 0.033Ha Permeable
 paving = 170m² = 0.017Ha Total Drained Area = 500m² Or 0.050Ha IL = 19.168mAOD Top of Crates =
 19.563mAOD Size Required: 6.0m x 12.0m x 0.4m deep crates with 95% voids...

<redacted_date>

Ref: C808/01/AS

Dear Chris,

Ref: Undertaking of Soakaway Tests at 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP

We refer to the undertaking of percolation tests at the above site, in support of the design of a possible
 soakaway system for the disposal of surface water from a proposed residential development.

Currently the parcel of land under investigation is the private garden to the residence at No 22 Market Lane,
 Bessacarr, Doncaster, ZZ29 6EP and comprises an extensive area of grassed soft landscaping.

The testing was undertaken at the locations shown below;

N

3.5 4.0 4.5 5.0 TP01 Trial Pit Complete 0.00 - 1.30m TP Volvo EC18 A Swinbourne Stable Inclination: 90° L = 1.40m W = 0.45m Trial Pit Sheet 1 of 1 Hole Type Easting Northing Ground Level (m) Scale TP 1:25 Project Name Project No. Start Date End Date 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr C808 <redacted_date>-<redacted_phone> Client Consultant Contractor J Roberts Design G&M Consulting Ltd Maxfield Excavation Remarks Method, Plant, Stability, Dimensions Logger Inst/ Backfill Results Legend Type/ Ref Depth (m) Samples and Tests Strata Water Levels Level (m) Depth (thickness) (m) Created using Pebble Geo TP01 Trial Pit Complete 0.00 - 1.30m TP Volvo EC18 A Swinbourne Stable Inclination: 90° L = 1.40m W = 0.45m Trial Pit PHOTO PAGE Hole Type Easting Northing Ground Level (m) Scale TP 1:25 Project Name Project No. Start Date End Date 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr C808 <redacted_date>-<redacted_phone> Client Consultant Contractor J Roberts Design G&M Consulting Ltd Maxfield Excavation Remarks Method, Plant, Stability, Dimensions Logger Created using Pebble Geo (0.50) 0.50 (0.45) 0.95 Description TOPSOIL: Dark brown clayey gravelly very silty fine and medium SAND. Gravel is subrounded fine to coarse of quartzite Firm becoming stiff, indistinctly fissured yellowish brown mottled light grey silty CLAY. Fissures randomly orientated End of Trial Pit at 0.95m 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0 TP02 Trial Pit Complete 0.00 - 0.95m TP Volvo EC18 A Swinbourne Stable Inclination: 90° L = 1.30m W = 0.45m Trial Pit Sheet 1 of 1 Hole Type Easting Northing Ground Level (m) Scale TP 1:25 Project Name Project No. Start Date End Date 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr C808 <redacted_date>-<redacted_phone> Client Consultant Contractor J Roberts Design G&M Consulting Ltd Maxfield Excavation Remarks Method, Plant, Stability, Dimensions Logger Inst/ Backfill Results Legend Type/ Ref Depth (m) Samples and Tests Strata Water Levels Level (m) Depth (thickness) (m) Created using Pebble Geo TP02 Trial Pit Complete 0.00 - 0.95m TP Volvo EC18 A Swinbourne Stable Inclination: 90° L = 1.30m W = 0.45m Trial Pit PHOTO PAGE Hole Type Easting Northing Ground Level (m) Scale TP 1:25 Project Name Project No. Start Date End Date 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr C808 <redacted_date>-<redacted_phone> Client Consultant Contractor J Roberts Design G&M Consulting Ltd Maxfield Excavation Remarks Method, Plant, Stability, Dimensions Logger Created using Pebble Geo (0.30) 0.30 (0.90) 1.20 (0.40) 1.60 Description MADE GROUND: Dark brown slightly clayey very gravelly fine to coarse SAND with a low cobble and boulder content. Gravel is subangular fine to coarse of brick, quartzite, limestone and tile. Cobbles and boulders are subangular of brick and concrete Firm becoming stiff, indistinctly fissured yellowish brown mottled light grey silty CLAY. Fissures randomly orientated Firm becoming stiff, indistinctly fissured dark grey mottled brown slightly gravelly silty CLAY with a low cobble content. Gravel is subrounded to rounded fine to coarse of quartzite. Cobbles are rounded of quartzite. Fissures randomly orientated End of Trial Pit at 1.60m 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0 TP03 Trial Pit Complete 0.00 - 1.60m TP Volvo EC18 A Swinbourne Stable Inclination: 90° L = 1.30m W = 0.45m Trial Pit Sheet 1 of 1 Hole Type Easting Northing Ground Level (m) Scale TP 1:25 Project Name Project No. Start Date End Date 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr C808 <redacted_date>-<redacted_phone> Client Consultant Contractor J Roberts Design G&M Consulting Ltd Maxfield Excavation Remarks Method, Plant, Stability, Dimensions Logger Inst/ Backfill Results Legend Type/ Ref Depth (m) Samples and Tests Strata Water Levels Level (m) Depth (thickness) (m) Created using Pebble Geo TP03 Trial Pit Complete 0.00 - 1.60m TP Volvo EC18 A Swinbourne Stable Inclination: 90° L = 1.30m W = 0.45m Trial Pit PHOTO PAGE Hole Type Easting Northing Ground Level (m) Scale TP 1:25 Project Name Project No. Start Date End Date 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr C808 <redacted_date>-<redacted_phone> Client Consultant Contractor J Roberts Design G&M Consulting Ltd Maxfield Excavation Remarks Method, Plant, Stability, Dimensions Logger Created using Pebble Geo (0.60) 0.60 (0.05) 0.65 Description TOPSOIL: Dark brown clayey gravelly very silty fine and medium SAND. Gravel is subrounded fine to coarse of quartzite Firm becoming stiff, indistinctly fissured yellowish brown mottled light grey silty CLAY. Fissures randomly orientated End of Trial Pit at 0.65m 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0 TP04 Trial Pit Complete 0.00 - 0.65m TP Volvo EC18 A Swinbourne Stable Inclination: 90° L = 0.40m W = 0.40m Trial Pit Sheet 1 of 1 Hole Type Easting Northing Ground Level (m) Scale TP 1:25 Project Name Project No. Start Date End Date 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr C808 <redacted_date>-<redacted_phone> Client Consultant Contractor J Roberts Design G&M Consulting Ltd Maxfield Excavation Remarks Method, Plant, Stability, Dimensions Logger Inst/ Backfill Results Legend Type/ Ref Depth (m) Samples and Tests Strata Water Levels Level (m) Depth (thickness) (m) Created using Pebble Geo TP04 Trial Pit Complete 0.00 - 0.65m TP Volvo EC18 A Swinbourne Stable Inclination: 90° L = 0.40m W = 0.40m Trial Pit PHOTO PAGE Hole Type Easting Northing Ground Level (m) Scale TP 1:25 Project Name Project No. Start Date End Date 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr C808 <redacted_date>-<redacted_phone> Client Consultant Contractor J Roberts Design G&M Consulting Ltd Maxfield Excavation Remarks Method, Plant, Stability, Dimensions Logger Created using Pebble Geo ATTACHMENT B FIELD DATA SHEETS Soakaway Test Results Sheet (Test No1) Job No C808 Site 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP Date <redacted_date> Head (m) Trial Pit Dimensions 0 527 Width (m) 0.45 1 527 Length (m) 1.4 2 Depth (m) 1.3 3 4 5 529 Infill depth (m) 10 Head of water (m) 15 25% Depth Value (m bgl) 20 75% Depth Value (m bgl) 25 30 45 25% Depth Value (Head m) 60 532 75% Depth Value (Head m) 75 90 105 120 534 AS 150 180 210 240

535 270 535 300 360 420 480 <redacted_date> Carried Out By TP No 3 Cumulative Time Standing Water (mm bgl) Cumulative Change (mm) Soakaway Test Results Sheet (Test No1) Job No C808 Site 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP Date <redacted_date> Head (m) Trial Pit Dimensions 0 198 Width (m) 0.4 1 201 Length (m) 0.4 2 204 Depth (m) 0.65 3 4 5 209 Infill depth (m) 10 209 Head of water (m) 15 25% Depth Value (m bgl) 20 75% Depth Value (m bgl) 25 30 45 213 25% Depth Value (Head m) 60 75% Depth Value (Head m) 75 90 214 105 120 215 AS 150 180 215 210 240 216 270 300 218 360 420 480
<redacted_date> Carried Out By TP No 4 Cumulative Time Standing Water (mm bgl) Cumulative Change (mm) J Roberts Design Ltd Page 1 11 The Point 4555 Bradmarsh Business Park 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, Bessacarr Rotherham, S60 1BP Date <redacted_date> Designed by LM File 1 IN 100Y+40%.MDX Checked by CAH Innovyze Network <redacted_date> STORM SEWER DESIGN by the Modified Rational Method Design Criteria for Storm ©<redacted_date>-<redacted_date> Innovyze Pipe Sizes STANDARD Manhole Sizes STANDARD FEH Rainfall Model Return Period (years) 2 FEH Rainfall Version <redacted_date> Site Location GB 461547 400711 SE 61547 00711 Data Type Point Maximum Rainfall (mm/hr) 50 Maximum Time of Concentration (mins) 30 Foul Sewage (l/s/ha) 0.000 Volumetric Runoff Coeff. 1.000 PIMP (%) 100 Add Flow / Climate Change (%) 0 Minimum Backdrop Height (m) 0.200 Maximum Backdrop Height (m) 1.500 Min Design Depth for Optimisation (m) 1.200 Min Vel for Auto Design only (m/s) 1.00 Min Slope for Optimisation (1:X) 500 Designed with Level Inverts Time Area Diagram for Storm Time (mins) Area (ha) Time (mins) Area (ha) 0-4 0.038 4-8 0.012 Total Area Contributing (ha) = 0.050 Total Pipe Volume (m³) = 0.750 Network Design Table for Storm PN Length (m) Fall (m) Slope (1:X) <redacted_url> (ha) T.E. (mins) Base Flow (l/s) k (mm) HYD SECT DIA (mm) Section Type Auto Design S1.<redacted_phone>...